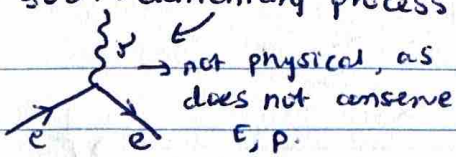
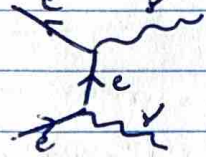


ELEMENTARY PARTICLE DYNAMICS:

→ QED: • elementary process



→ Pair annihilation ($e^+e^- \rightarrow 2\gamma$)

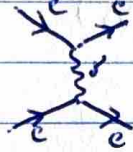


→ No photon-photon vertex

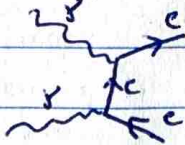
→ higher order diagrams: each vertex adds factor of $\alpha_e = 1/137$ so QED is perturbative, we can neglect higher order.

• Other processes:

→ Møller scattering (e^-e^- scattering)



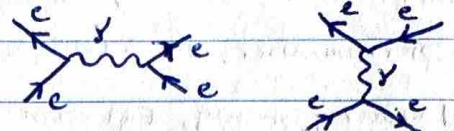
→ Pair production ($2\gamma \rightarrow e^+e^-$)



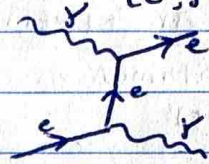
→ Virtual particles: do not lie on mass shell. (internal lines)

→ Real particles: lie on mass shell (external lines)

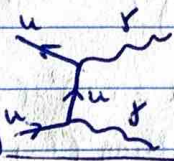
→ Bhabha scattering (e^+e^- scattering)



→ Compton scattering ($e^-\gamma$ scattering)

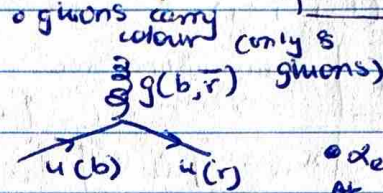
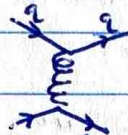
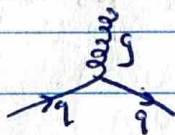


→ Quark QED: e.g. pion decay $\pi^0 \rightarrow 2\gamma$ [π^0 decays electromagnetically, faster than $\pi^\pm$, which decay by weak force]



→ QCD: • elementary process:

• force between quarks.



• gluons carry colour (only 8 gluons)

• gluons can couple: 3-gluon, 4-gluon vertices.

• $\alpha_s \rightarrow$ running constant, greater than 1 at larger distances (nuclear), (lower energies $\rightarrow$ non-perturbative QCD), smaller than 1 at smaller distances (inside hadrons), higher energies $\rightarrow$ perturbative QCD $\rightarrow$ quarks are asymptotically free.

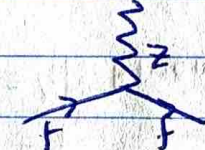
• In QCD, also vacuum polarisation, but also have gluon loops, which drives α_s down at short dist. while quark loops drives it up at short dist. Behaviour depends on $a \equiv 2f - 11n/3 = \text{no. of colours}$. if a is +ve, then α_s increases with short dist, and vice versa. ($f=6, n=3 \Rightarrow a=-21 \Rightarrow \alpha_s$ decreases at short dist.)

• all naturally occurring particles are colour singlets.

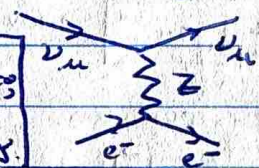
• α_e also not constant. At large dist. charge q screened by vacuum polarization. at very small dist. unscreened charge can be measured.

→ Weak interactions:

• Neutral: elementary process.



→ ν, e^- scattering:

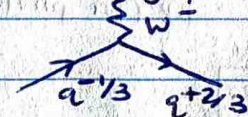


→ Z can mediate any process that the photon mediates: e.g. e^-e^- scattering, but photon dominates that process $\rightarrow$ minute correction to EM processes attributable to Z

$\therefore$ for pure neutral weak reaction, we must use ν .

→ W.I. bosonic vertices: $3W, 1Z$, $4W, 2W, 2Z$, W also couples to g .

→ quarks: elementary process:



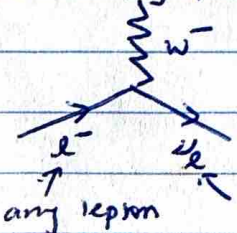
→ weak force couples (u, d') (c, s') (t, b')

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

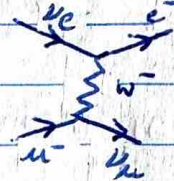
→ Kobayashi-Maskawa matrix

NOT unit matrix $\rightarrow$ allows u to couple to $s \rightarrow$ strangeness changing weak interactions.

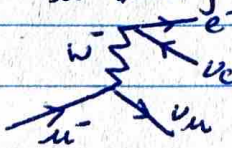
• charged: $\rightarrow$ leptons. elementary process



→ $u + \nu_e \rightarrow e^- + \nu_u$



→ difficult process to observe, but same as μ^- decay:

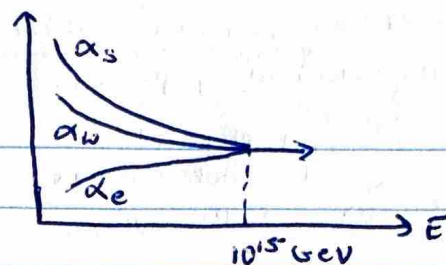


any lepton corresponding

→ Decays + conservation laws:

- charge: All 3 conserve
- colour: Strong conserves (EM, W, I don't affect)
- Baryon number: All 3 conserve (quarks: $+1/3$, All mesons have baryon # 0, antiquarks: $-1/3$)
- lepton number: All 3 conserve
- Flavour: Strong, EM conserve
Weak does not conserve

→ coupling constant behaviour:



RELATIVISTIC KINEMATICS

• Lorentz transformations:

$$\begin{cases} \text{i. } x' = \gamma(x - vt) \\ \text{ii. } y' = y \\ \text{iii. } z' = z \\ \text{iv. } t' = \gamma(t - \frac{v}{c^2}x) \end{cases}$$

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}$$

• Inverse transforms:

$$\begin{cases} \text{i. } x = \gamma(x' + vt') \\ \text{ii. } y = y' \\ \text{iii. } z = z' \\ \text{iv. } t = \gamma(t' + \frac{v}{c^2}x') \end{cases}$$

• Four vectors:

↳ position-time: $x^\mu \rightarrow$ contravariant

$$\begin{cases} x^0 = ct \\ x^1 = x \\ x^2 = y \\ x^3 = z \end{cases} \rightarrow \text{Lorentz transforms} \begin{cases} x'^0 = \gamma(x^0 - \beta x^1) \\ x'^1 = \gamma(x^1 - \beta x^0) \\ x'^2 = x^2 \\ x'^3 = x^3 \end{cases}$$

$$\beta \equiv \frac{v}{c}$$

$$\Rightarrow x'^\mu = \Lambda^\mu_\nu x^\nu \rightarrow \text{summation over } \nu$$

$$\Lambda = \begin{pmatrix} \gamma - \gamma\beta & 0 & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

↳ Invariant: $I \equiv (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$

↳ Metric: $g = \text{diag}(1, -1, -1, -1)$

$\Rightarrow I = g_{\mu\nu} x^\mu x^\nu \rightarrow$ summation over μ, ν

$$[g_{\mu\nu} = g^{\mu\nu}]$$

↳ covariant 4-vector: $x_\mu = g_{\mu\nu} x^\nu$

$\Rightarrow x_0 = x^0, x_1 = -x^1, x_2 = -x^2, x_3 = -x^3$

$\Rightarrow I = x_\mu x^\mu = x_\mu x^\mu$

↳ given two four-vectors a^μ, b^μ

$\Rightarrow a_\mu b^\mu = a^\mu b_\mu = a^0 b^0 - a^1 b^1 - a^2 b^2 - a^3 b^3$

$a \cdot b \rightarrow$ invariant

$$a \cdot b = a^0 b^0 - \vec{a} \cdot \vec{b}$$

$a_\mu a^\mu = a^2 \rightarrow$ if $a^2 > 0$: $a^\mu \rightarrow$ timelike
if $a^2 < 0$: $a^\mu \rightarrow$ spacelike
if $a^2 = 0$: $a^\mu \rightarrow$ lightlike

↳ Proper time: $d\tau \equiv \frac{dt}{\gamma} \rightarrow$ time, on particles watch

(in particle's reference frame) $\rightarrow$ invariant

↳ proper velocity: $\eta \equiv \frac{dx}{d\tau} = \gamma v \rightarrow \eta^\mu = \frac{dx^\mu}{d\tau}$

$\Rightarrow \eta^\mu = \gamma(c, v_x, v_y, v_z)$

$\Rightarrow \eta^\mu \eta_\mu = c^2 \rightarrow$ invariant

↳ Relativistic momentum: $p \equiv m\eta \Rightarrow p^\mu = m\eta^\mu$

$$\vec{p} = \gamma m \vec{v} = \frac{m \vec{v}}{\sqrt{1 - v^2/c^2}}, \quad p^0 = \gamma mc$$

↳ Relativistic energy: $E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$

$\Rightarrow p^\mu = (\frac{E}{c}, p_x, p_y, p_z)$

$\Rightarrow p^\mu p_\mu = \frac{E^2}{c^2} - \vec{p}^2 = m^2 c^2$

↳ Rest energy: $E \equiv mc^2$

↳ massless particles:

$v = c, \quad E = |\vec{p}|c$
 $= h\nu$ (for photons)

$$\vec{v} = \frac{\vec{p}c^2}{E}$$

SYMMETRIES

→ Properties of a group:

- Closure: the combination of two elements by the binary operator gives another member of the group.
- Identity: There is an identity element, which when combined with any group element gives back that element.
- Inverse: every element has an inverse element, which combined w/ it to give the identity.
- Associativity: $(123) = (12)3$

→ Abelian group: commuting elements.

→ $U(n)$: group of unitary matrices of dim. n [$U^{-1} = U^\dagger$]→ $SU(n)$: $U(n)$ with $\det. = 1$ → $O(n)$: real unitary matrices.→ $SO(n)$: $O(n)$ with $\det. = 1$ → Addition of angular momentum (J_1, J_2)

z components add: $m_{\text{tot}} = m_1 + m_2$

magnitudes do not: $j_{\text{tot}} = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, (j_1 + j_2) - 1, (j_1 + j_2) \rightarrow$ get all possible values from $|j_1 - j_2|$ to $(j_1 + j_2)$

→ Flavour symmetries:

↳ isospin $\rightarrow$ analogous to spin

$$p = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left| \frac{1}{2}, \frac{1}{2} \right\rangle \quad n = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

(isospin up) (isospin down)

↳ strong interactions are invariant under rotations in isospin space:

invariant under $SU(2)$ rotations
multiplets are assigned an isospin I ,
each member gets a particular I_3

$$\pi \rightarrow I=1: \pi^+ = |1, 1\rangle, \pi^0 = |1, 0\rangle, \pi^- = |1, -1\rangle$$

$$\Lambda \rightarrow I=0: \Lambda = |0, 0\rangle$$

$$\Delta, I = \frac{3}{2}: \Delta^{++} = \left| \frac{3}{2}, \frac{3}{2} \right\rangle, \Delta^+ = \left| \frac{3}{2}, \frac{1}{2} \right\rangle$$

$$\Delta^0 = \left| \frac{3}{2}, \frac{1}{2} \right\rangle, \Delta^- = \left| \frac{3}{2}, -\frac{1}{2} \right\rangle$$

multiplicity (number of members in multiplet) = $2I + 1$

Gell-Mann - Nishijima formula:

$$Q = I_3 + \frac{1}{2}(A + S)$$

Q : charge

A : baryon number

S : strangeness

↓
for hadrons
made of u, d, s .

→ Angular momentum:

• orbital: $L \rightarrow L^2 = \ell(\ell+1)\hbar^2$

$$\ell = 0, 1, 2, 3, \dots$$

$$L_z = m_\ell \hbar$$

$$m_\ell = -\ell, -\ell+1, \dots, -1, 0, 1, \dots, \ell-1, \ell$$

$$\ell = 0, 1, 2, \dots, (2\ell+1)$$

• spin: $S \rightarrow S^2 = s(s+1)\hbar^2$

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, \dots$$

$$S_z = m_s \hbar$$

$$m_s = -s, -s+1, \dots, s-1, s$$

Bosons

(integer spin)

$s=0$	$s=1$	$s=1/2$	$s=3/2$
pseudo-scalar nonet (mesons)	force mediators vector nonet (mesons)	quarks + leptons Baryon octet	Baryon decuplet

Fermions

(half-integer spin)

e.g.: $s=1, \ell=3$

$$\Rightarrow j=2, j=3, j=4$$

↳ quark model:

$$u = \left| \frac{1}{2}, \frac{1}{2} \right\rangle, d = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, s = |0, 0\rangle$$

$$p, n \rightarrow \text{triplet} + \text{singlet}: 2 \otimes 2 = 3 \oplus 1$$

$$|1, -1\rangle = pp, |1, 0\rangle = \frac{1}{\sqrt{2}}(pn + np), |1, 1\rangle = nn$$

↳ symmetric

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(pn - np) \rightarrow \text{antisymmetric}$$

only bound state of p, n is deuteron
 $d = |0, 0\rangle$

→ discrete symmetries:

↳ Parity: W violates parity maximally

helicity: m_S / S : $+1 \rightarrow$ right handed
(not Lorentz invariant) $-1 \rightarrow$ left handed.

↳ helicity of neutrinos: Lorentz invariant
as they are practically massless

All neutrinos are left handed

All anti-neutrinos are right handed

The mirror image of a neutrino
does not exist.

→ Parity:

Parity → inversion

$$P(x, y, z) = (-x, -y, -z) \quad \therefore \text{Parity group}$$

$$P^2 = I \Rightarrow \text{eigenvalues of } P = \pm 1$$

Scalar: $P(s) = s$

Pseudoscalar: $P(p) = -p$

vector $P(v) = -v$

axial vector $P(a) = a$
(pseudo)

→ Strong + EM
conserve parity
WI violates parity.

• Hactions are eigenstates of P.

→ parity of fermion → opp. of antifermion

→ parity of boson → same as antiboson.

→ quarks: $P = +1$

antiquarks: $P = -1$

→ Parity is multiplicative:

∴ Baryon octet + decuplet: $P = (+1)^3 = +1$

meson nonets: $P = (+1)(-1) = -1$

→ For an excited state of two particles: extra factor

∴ mesons: $P = (-1)^{L+1}$

of $(-1)^L$
L = angular momentum.

→ photon: $P = -1$

→ charge conjugation:

$$C|p\rangle = |\bar{p}\rangle \rightarrow \text{converts particle into antiparticle.}$$

→ photon: $C = -1$

$$C^2 = I \Rightarrow \text{eigenvalues of } C = \pm 1$$

→ for system of spin-1/2 particle + antiparticle.

only particles that are their own antiparticles are eigenstates of C.

$$C = (-1)^{L+S}$$

→ C is multiplicative

• pseudoscalars ($L=0, S=0$) → $C = +1$

• vectors ($L=0, S=1$) → $C = -1$

→ Strong + EM conserve C

WI violates C.

→ G parity: $G = CR_2$, $R_2 = e^{i\pi I_2} \rightarrow 180^\circ$ rotation

→ $G = (-1)^I C$ conserved by strong, not EM, WI. about I_2 axis. converting $I_3 \rightarrow -I_3$ for neutral member

→ CP: violated by WI.

neutral kaons: $K^0, \bar{K}^0$
 $S=+1, S=-1$

$$P|K^0\rangle = -|\bar{K}^0\rangle, P|\bar{K}^0\rangle = -|K^0\rangle \text{ (pseudoscalar)}$$

$$C|K^0\rangle = |\bar{K}^0\rangle, C|\bar{K}^0\rangle = |K^0\rangle \Rightarrow CP|K^0\rangle = -|\bar{K}^0\rangle$$

∴ eigenstates of CP

$$CP|\bar{K}^0\rangle = -|K^0\rangle$$

$$|K_1\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle), |K_2\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle) \Rightarrow CP|K_1\rangle = |K_1\rangle, CP|K_2\rangle = -|K_2\rangle$$

∴ If WI conserves CP → $K_1 \rightarrow$ decay to 2π ($CP=+1$), $K_2 \rightarrow$ decay to 3π ($CP=-1$)

$K_1 \rightarrow 2\pi$ happens much faster shorter lifetime $P=+1, C=+1$

$P=-1, C=+1$

So if we start w/ $|K^0\rangle \rightarrow$ the K_1 component must decay away quickly, and away from the source we must observe only 3π events.

BUT, 2π events were observed away from the source ∴ CP is violated.

BOUND STATES

→ Light quark mesons: (u, d, s)

ground state: $l=0$

→ spin state: parallel ($s=1$) → vector nonet

antiparallel ($s=0$) → pseudoscalar nonet

→ u and d form isospin doublet:

$$u = |\frac{1}{2}, \frac{1}{2}\rangle, \quad d = |\frac{1}{2}, -\frac{1}{2}\rangle$$

$$\bar{d} = -|\frac{1}{2}, \frac{1}{2}\rangle, \quad \bar{u} = |\frac{1}{2}, -\frac{1}{2}\rangle$$

Combine two particles with $I=1/2$

$$\rightarrow \text{isotriplet: } \begin{cases} |1, 1\rangle = -u\bar{d} \\ |1, 0\rangle = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}) \\ |1, -1\rangle = d\bar{u} \end{cases} \rightarrow \begin{cases} \pi^+, \rho^+ \\ \pi^0, \rho^0 \\ \pi^-, \rho^- \end{cases}$$

$$\rightarrow \text{isosinglet: } |0, 0\rangle = \frac{1}{\sqrt{2}}(u\bar{u} + d\bar{d}) \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{ gives us } \eta, \eta', \omega, \phi$$

→ $Su(2)$ gives another isosinglet: $|0, 0\rangle = s\bar{s}$

$$\eta = \frac{1}{\sqrt{6}}(u\bar{u} + d\bar{d} + 2s\bar{s}) \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{ pseudoscalar}$$

$$\eta' = \frac{1}{\sqrt{3}}(u\bar{u} + d\bar{d} + s\bar{s})$$

→ singlet under $Su(3)$

$$\omega = \frac{1}{\sqrt{2}}(u\bar{u} + d\bar{d}) \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{ vector}$$

$$\phi = s\bar{s}$$

not a singlet under $Su(3)$

∴ $Su(3)$ is not a good symmetry for vector nonet.

$$\rightarrow \text{quarks: } 3 \quad \left. \begin{array}{l} \\ \end{array} \right\} 3 \otimes \bar{3} = 8 \oplus 1$$

antiquarks: $\bar{3}$

All particles in a supermultiplet do not have the same mass $\Rightarrow Su(3)$ is not a good symmetry $\rightarrow$ broken

→ meson mass:

$$m_u = m_d = 308 \text{ MeV}/c^2$$

$$m_s = 483 \text{ MeV}/c^2$$

$$A = \left(\frac{2m_u}{\hbar^2}\right) 159 \text{ MeV}/c^2$$

$$M(\text{meson}) = m_1 + m_2 + A \frac{(S_1 \cdot S_2)}{m_1 m_2}$$

$$\rightarrow S_1 \cdot S_2 = \begin{cases} \frac{1}{4} \hbar^2 \rightarrow s=1 (\text{vector}) \\ -\frac{3}{4} \hbar^2 \rightarrow s=0 (\text{pseudoscalar}) \end{cases}$$

→ Baryons

• wavefunctions: 2 orbital angular momenta (l, l')

ground state: $l=l'=0$

∴ J comes totally from S . $\rightarrow S = 3/2$ or $1/2$

→ 3-quark spin states:

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2 \rightarrow \text{symmetric quadruplet, 2 doublets of mixed symmetry. (pg 182)}$$

$$|\psi_1\rangle \rightarrow \text{antisymmetric in } 1, 2$$

$$|\psi_2\rangle \rightarrow \text{antisymmetric in } 1, 3$$

$$|\psi_3\rangle = |\psi_1\rangle + |\psi_2\rangle$$

→ Baryons: fermions, antisymmetric wavefunction

$\Rightarrow \psi(1,2) = -\psi(2,1) \Rightarrow$ Pauli exclusion principle.

$$\psi(\text{baryon}) = \psi(\text{space}) \psi(\text{spin}) \psi(\text{flavour}) \psi(\text{colour})$$

symmetric combination must be symmetric. (nature always chooses antisymmetric singlet)

→ colour, flavour: $3 \otimes 3 \otimes 3 \rightarrow 10 \oplus 8 \oplus 8 \oplus 1$

$\psi_{13} = \psi_{12} + \psi_{23}$

symmetric
decuplet
flavour states

mixed
symmetry
octets

antisymmetric singlet.

$\psi(\text{colour}) = \frac{1}{\sqrt{6}} (rgb - rbg + gbr - grb + brg - bgr)$

ψ_5 : completely symmetric

ψ_{12} : antisymmetric in 1,2

ψ_{23} : antisymmetric in 2,3

$\psi(\text{baryon decuplet}) = \psi_5(\text{spin}) \psi_5(\text{flavour})$

$\psi(\text{baryon octet}) = \frac{\sqrt{2}}{3} (\psi_{12}(\text{spin}) \psi_{12}(\text{flavour}) + \psi_{23}(\text{spin}) \psi_{23}(\text{flavour}) + \psi_{13}(\text{spin}) \psi_{13}(\text{flavour}))$

• Magnetic moments: octet baryons:

$\mu = \mu_1 + \mu_2 + \mu_3 \rightarrow$ for particle of spin $1/2$, charge q , mass m , $\mu = \frac{q\hbar}{2mc} \rightarrow \mu_z$

$\mu_u = \frac{2}{3} \frac{e\hbar}{2m_u c}$, $\mu_d = -\frac{1}{3} \frac{e\hbar}{2m_d c}$, $\mu_s = -\frac{1}{3} \frac{e\hbar}{2m_s c}$

∴ Magnetic moment of baryon:

$m_u = m_d = 336 \text{ MeV}/c^2$

$m_s = 538 \text{ MeV}/c^2$

$$\mu_B = \langle B \uparrow | (\mu_1 + \mu_2 + \mu_3)_z | B \uparrow \rangle$$

$$= \frac{2}{\hbar} \sum_i \langle B \uparrow | (\mu_i S_{iz}) | B \uparrow \rangle$$

 ↑ wavefunction of baryon (spin, flavour)

• Masses:

$M(\text{baryon}) = m_1 + m_2 + m_3 + A' \left[\frac{S_1 \cdot S_2}{m_1 m_2} + \frac{S_1 \cdot S_3}{m_1 m_3} + \frac{S_2 \cdot S_3}{m_2 m_3} \right]$

$m_u = m_d = 336 \frac{\text{MeV}}{c^2}$
 $m_s = 538 \frac{\text{MeV}}{c^2}$

$A' = \left(\frac{2m_u}{\hbar^2} \right)^2 50 \frac{\text{MeV}}{c^2}$

→ when all 3 quark masses are equal: $m_1 = m_2 = m_3$

(baryons made of u, d) → $S_1 \cdot S_2 + S_1 \cdot S_3 + S_2 \cdot S_3 = \begin{cases} \frac{3}{4} \hbar^2 & j = \frac{3}{2} (\text{decuplet}) \\ -\frac{3}{4} \hbar^2 & j = \frac{1}{2} (\text{octet}) \end{cases}$

∴ $M_N = M_p = 3m_u - \frac{3}{4} \frac{\hbar^2}{m_u^2} A'$

$M_\Delta = 3m_u + \frac{3}{4} \frac{\hbar^2}{m_u^2} A'$

$M_{\Lambda^-} = 3m_s + \frac{3}{4} \frac{\hbar^2}{m_s^2} A'$

→ $\Sigma \rightarrow$ triplet → u, d quarks make isospin 1.

∴ for spin, flavour wavefunction to be symmetric under exchange of u, d , they must combine to make spin 1

⇒ $(S_u + S_d)^2 = S_u^2 + S_d^2 + 2S_u \cdot S_d = 2\hbar^2$
 ⇒ $S_u \cdot S_d = \frac{\hbar^2}{4}$

→ Baryon decuplet: All spins are parallel (every spin combines to make spin 1)

⇒ $S_1 \cdot S_2 = S_1 \cdot S_3 = S_2 \cdot S_3 = \frac{\hbar^2}{4}$

⇒ $M_{\Sigma^0} = 2m_u + m_s + \frac{\hbar^2}{4} A' \left(\frac{1}{m_u^2} + \frac{2}{m_u m_s} \right)$

$M_{\Xi^0} = m_u + 2m_s + \frac{\hbar^2}{4} A' \left(\frac{2}{m_u m_s} + \frac{1}{m_s^2} \right)$

→ $\Lambda \rightarrow$ singlet → $u, d \rightarrow$ isospin 0 → spin 0

⇒ $(S_u + S_d)^2 = 0 \Rightarrow S_u \cdot S_d = -\frac{3}{4} \hbar^2$

$M_\Lambda = 2m_u + m_s - \frac{3}{4} \frac{\hbar^2}{m_u^2} A'$

⇒ $M_{\Xi^-} = 2m_u + m_s + A' \left[\frac{S_u \cdot S_d}{m_u m_d} + \frac{(S_1 \cdot S_2 + S_1 \cdot S_3 + S_2 \cdot S_3 + S_u \cdot S_d)}{m_u m_s} \right]$

But $S_1 \cdot S_2 + S_1 \cdot S_3 + S_2 \cdot S_3 = -\frac{3}{4} \hbar^2 \rightarrow$ octet

⇒ $M_{\Xi^-} = 2m_u + m_s + \frac{\hbar^2}{4} A' \left[\frac{1}{m_u^2} - \frac{4}{m_u m_s} \right]$

→ Ξ^- : $M_{\Xi^-} = 2m_s + m_u + A' \left[\frac{S_1 \cdot S_2}{m_s^2} + \frac{(S_1 + S_2) \cdot S_u}{m_s m_u} \right]$

(1,2 → s quark numbers)

1,2 → both s quarks ∴ flavour wavef. is symm in 1,2

1,2 → spin wavef. must be symm. in 1,2 → $S_1, S_2 = 1$

⇒ $S_1 \cdot S_2 = \frac{\hbar^2}{4}$, $S = (S_1 + S_2) + S_u \Rightarrow (S_1 + S_2) \cdot S_u = -\hbar^2$ (squaring)

⇒ $M_{\Xi^-} = 2m_s + m_u + \frac{\hbar^2}{4} A' \left(\frac{1}{m_s^2} - \frac{4}{m_u m_s} \right)$